Ex 12 from the exam on 04/02/2026, row 1

A critical component used to implement an IT service is characterized by an MTTF = 640 hours and an MTTR = 8 hours. The availability of the service directly depends on the availability of this component, which represents a fundamental building block of the service architecture. The service can be implemented using two alternative configurations:

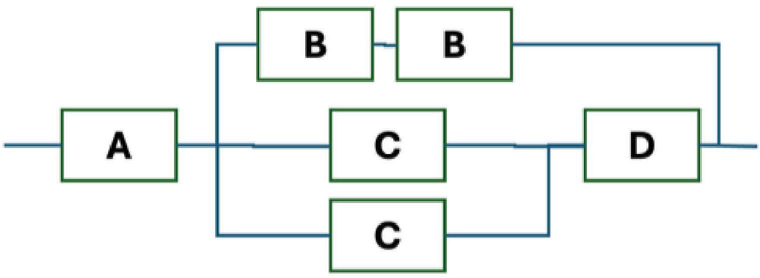
- Configuration A (single): the service depends on a single component.
- Configuration B (active redundancy): the service is implemented using two identical components and it is available if at least one of the two components is operational.

Given that service unavailability costs €800 per hour, estimate the expected annual cost savings (over a period of 365 days) in terms of downtime obtained by adopting Configuration B instead of Configuration A.

[85450 €]

Ex 12 from the exam on 12/06/2025, row 4

Suppose we have a computer system composed of 6 different components, and designed to have an RBD as shown in the image below. The four types of components (A, B, C, and D) have different reliability characteristics. We know that the availability of the components B, C, and D are respectively $Av_A = Av_B = 0.8$, $Av_C = 0.7$, and $Av_D = 0.9$. What is the availability of the entire system? Use always at least 4 decimal digits for each calculation.



[0.748]

Ex 12 from the exam on 04/07/2025, row 3

A datacenter hosts a critical service that relies on a system composed of three components:

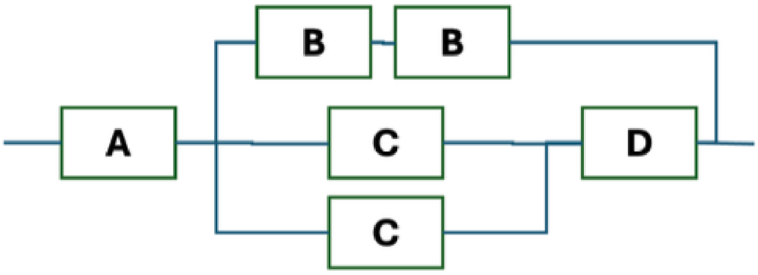
- Component A is a central control unit (e.g., a network controller or orchestrator) that is mandatory for system operation. If A fails, the service becomes unavailable.
- Components B1 and B2 are redundant processing units (e.g., heterogeneous application servers) configured such that only one needs to be operational for the system to function correctly.

We know that the reliability of A after 2 years is 0.8, while its MTTR (Mean Time to Repair) is 14 days. Moreover, we know the Availability of B1 and B2, which are respectively 0.75 and 0.85. Compute the availability of component A and compute the overall system availability.

[0.9957, 0.9584]

Ex 11 from the exam on 12/06/2024, row 1

Suppose we have a computer system composed of 6 different components, and designed to have an RBD as shown in the image below. The four types of components (A, B, C, and D) have different reliability characteristics. We know that after 2 years the reliability of components B, C, and D are respectively $R_B(2y) = 0.8$, $R_C(2y) = 0.6$, and $R_D(2y) = 0.9$. What should be the MTTF for component A, if we want to have a Reliability of the whole system after 2 years equal to $R_{sys}(2y) = 0.82$? Use always at least 3 decimal digits for each calculation.



[18.777 y]

Ex 12 from the exam on 11/09/2025, row 5

A datacenter operates a multi-tier web application composed of the following components both needed to process a request. Component A is the front-end layer composed of the load balancer (A1) and the front-end server (A2) required, both required for all incoming traffic. If A fails, no request can be processed. Component B is the application layer, composed of three identical and independent compute nodes (B1, B2, B3). Component B continues to operate as long as at least one compute node is available. We know that the availability of the load balancer and of the front-end server is 0.96 in both cases (A1 and A2). The availability of each compute node of the application layer (B1, B2, B3) is 0.65. Compute the overall system availability.

[0.882]

Ex 13 from the exam on 10/07/2024, row 6

During the procurement of a server for an important scientific calculation, 3 different solutions have been offered.

- Server A allows to complete the target calculation in 400 hours, it has a MTTF of 1600 hours, and a MTTR of 3 hours;
- Server B allows to complete the target calculation in 500 hours, it has a MTTF of 1750 hours, and a MTTR of 4 hours.
- Server C allows to complete the target calculation in 600 hours, and it has a MTTR of 5 hours.

We know that the decision on which solution to buy depends on which server has the higher probability of completing the calculation before failure, once the calculation it is started. What should be the minimum MTTF for Server C to be selected as the system to buy? Use at least 4 decimal digit for each intermediate calculation.

[2400 h]

Ex 11 from exam on 13/01/2025, row 3

A scientific computation that needs to be carried out within the PoliMi data center uses a server composed of 2 CPUs and 4 GPUs. Knowing that:

- The computation takes 7 days to complete,
- The computation requires at least one CPU and all GPUs within the server to be operational to complete successfully;
- $MTTF_{CPU} = 180$ days and $MTTF_{GPU} = 120$ days.

How many parallel instances of the computation must be launched to ensure a probability higher than 98% that at least one computation produces results successfully? Notes: (i) Use at least 4 decimal places for all intermediate calculations. (ii) All other components of the server can be considered ideal.

[3]

Ex 12 from exam on 03/02/2025, row 5

A company uses a computing system composed of three servers: one front-end server and two back-end servers. The system remains operational even if one of the two back-end servers fails. In all other cases, the system is considered offline. All servers have a Mean Time To Failure (MTTF) of 8 months. (i) Compute the probability that at least one of the three servers fails within the first 40 days, and (ii) compute the overall reliability of the computing system after 40 days. *Notes: Use at least 4 decimal places for each intermediate calculation.*

[0.3934, 0.8265]